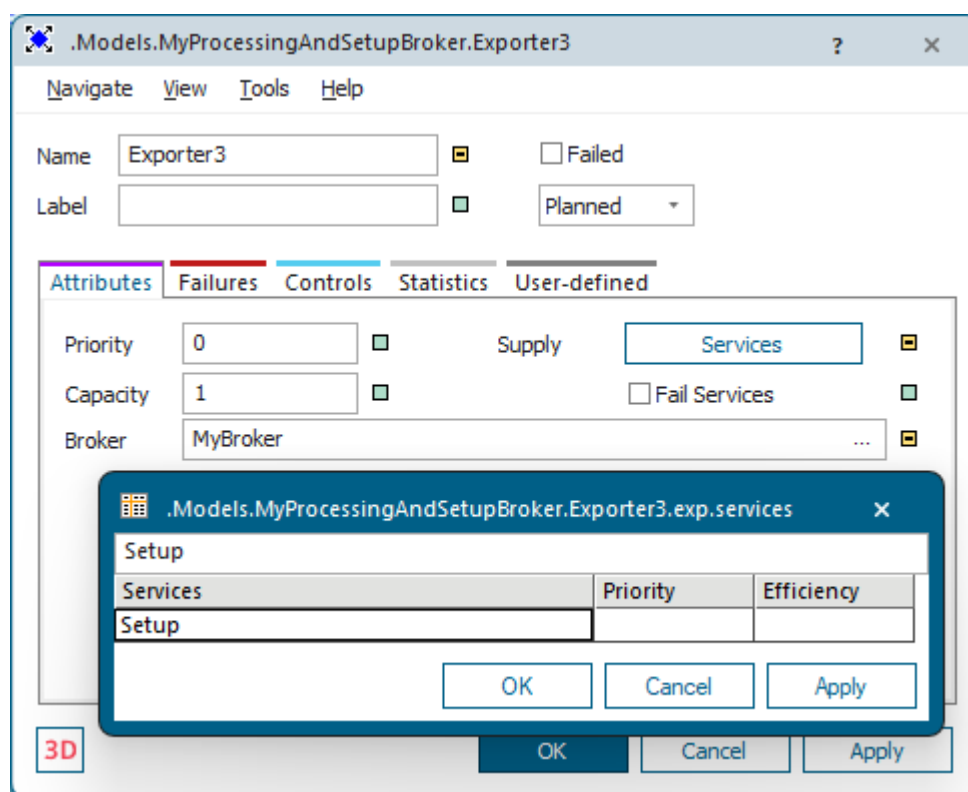




- Finally, open the *EventController* and click **Start/Stop Simulation**. As the *Exporter* can only work at one station at a time, and because the stations *Station1* and *Station3* share *Exporter1*, short standstills occur. *Station1* has to wait until *Station3* releases *ExporterJob1*, and thus is blocked. *Station3* in turn has to wait until *Station1* releases *Exporter1*, and thus is **Waiting**. All stations receive the service **Setup** from *Exporter3*, which does not cause a problem as it is needed once only for each station.

You can also click the tab **Statistics** and view the most important data, which the object collected.

Back to [Model Workers with Importer, Broker, and Exporter](#)

Modeling a Shift System

Model a Shift System

When creating your simulation model, you will, more than likely, also have to integrate the shift system that applies to your plant. You can quickly and easily define these shifts with the object *ShiftCalendar*.

The **ShiftCalendar**  can control the working hours as well as the paused times, the planned times, and the unplanned times of these **material flow objects**: *Frame* , *Station* , *ParallelStation* , *AssemblyStation* , *DismantleStation* , *Buffer* , *PlaceBuffer* , *Store* , *Sorter* , *Conveyor* , *AngularConverter* , *Turntable* , *Turnplate*, *Track* , *TwoLaneTrack* , *Source* , *Drain* , *WorkerPool* , and *Exporter* .

Naturally, you can also control *Frames* in which you modeled components of your overall plant, with the *ShiftCalendar*. Here you use *Methods* as unplanned time and paused time controls of that *Frame* to propagate the shift settings to the objects, which you inserted into this *Frame*.

The *ShiftCalendar* can control the working hours as well as the paused times, the planned times, and the unplanned times of these **fluid objects**: *FluidSource* , *Mixer* , *Portioner* , and *DePortioner* .

You can use the method **schedule** to make the *ShiftCalendar* set the date and time to start or to finish the production process.

Starting and Ending a Shift

When a *ShiftCalendar* starts a shift, it sets the attribute **Unplanned** of the material flow objects and of any of the *Frames* at controls to false, i.e., it deactivates unplanned times. If need be, you can program an **Unplanned Control** that executes other actions, which your modeling situation requires.

The statistics of the material flow objects then starts to collect the unplanned time. This is the time when a resource is not planned to work.

When you define shifts that are active from 6 o'clock in the morning until 10 o'clock at night, for example, the **planned time** lasts from 6 o'clock to 22 o'clock. The **unplanned time** lasts from 22 o'clock to 6 o'clock the following morning.

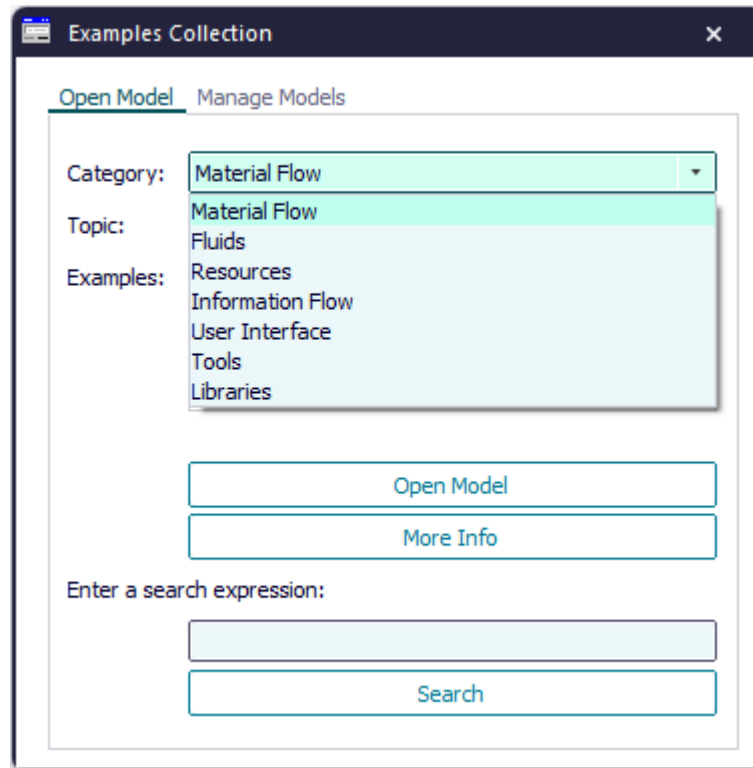
When a *ShiftCalendar* finishes a shift, it sets the attribute *Unplanned* of the material flow object and *Frames* it controls to true, i.e., it activates unplanned times. Then, the objects stop processing the current part and release all services.

Starting and Ending a Break

When a *ShiftCalendar* starts a break, it sets the attribute **Pause** of the material flow objects and of the *Frames* it controls to true, i.e., it activates pauses. For the material objects activating a pause also affects statistics. For *Frames Plant Simulation* only changes the state of the attribute. If need be, you can program a **Pause Control** for your *Frames* that executes the actions your modeling situation requires.

When a *ShiftCalendar* ends a break, it sets the attribute *Pause* of the material flow object and *Frames* it controls to false, i.e., it deactivates pauses. For the *Frames* you can react to the end of a pause in a **pause control** that executes the actions your modeling situation requires.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



Go to [Define Shifts with the ShiftCalendar](#)

Go to [Pause Material Flow Objects and Frames](#)

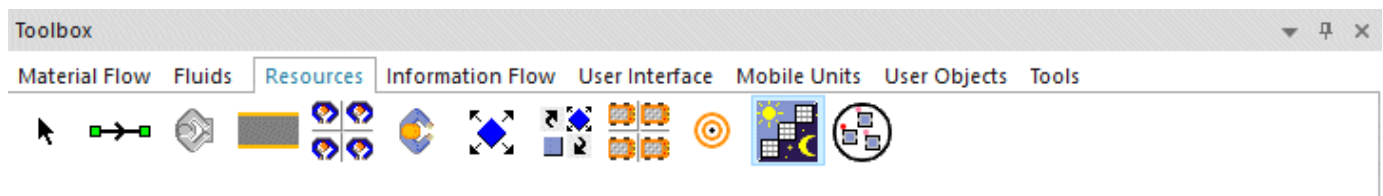
Back to [Modeling the Material Flow](#)

Defining Shifts with the ShiftCalendar

Define Shifts with the ShiftCalendar

You can define the settings for the entire shift system in the *ShiftCalendar*.

You can insert the *ShiftCalendar* into your simulation model from the folder **Resources** in the *Class Library* or from the toolbar **Resources** in the *Toolbox*.



You can define the settings for the entire shift system in the [ShiftCalendar](#) .

You can:

- [Specify the Names of the Shifts, the Respective Times and Days](#)
- [Specify Times During Which the Plant Works Part of the Time](#)
- [Specify the Stations Which the ShiftCalendar Controls](#)
- [Schedule Date and Time to Start or Finish the Production Process](#)

Once you have entered all the settings for the shifts, you have to select the check box **Active**, so that *Plant Simulation* uses the shifts in your simulation model.

Compare [Pause Material Flow Objects and Frames](#)

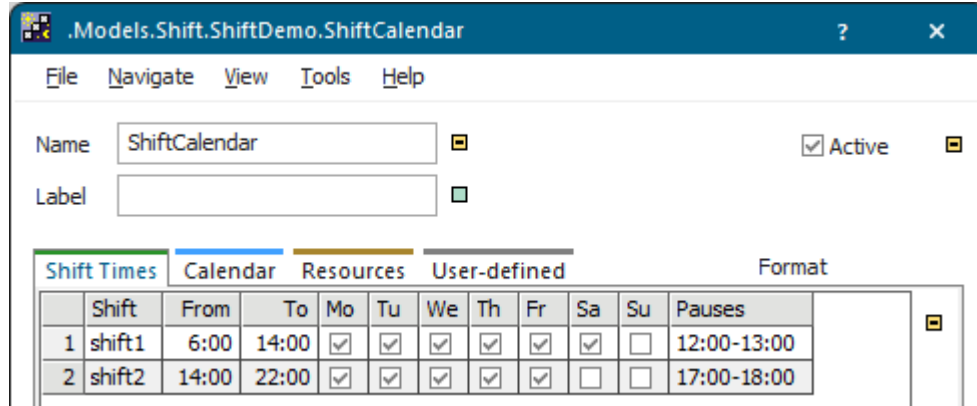
Back to [Model a Shift System](#)

Specify the Names of the Shifts, the Respective Times and Days

You can specify the names of the shifts, the respective times and days.

Before you can type in data, click the **Inheritance** check box so that it looks like this .

Type the data pertaining to one shift into the cells of one row.



Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1 shift1	6:00	14:00	☒	☒	☒	☒	☒	☒	☐	12:00-13:00
2 shift2	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	17:00-18:00

- Type the names of the different shifts into the cells below **Shift**. You might, for example, enter **Morning shift**, **Day shift**, **Evening shift**, **Graveyard shift**, etc.
- Type the times at which the shifts start into the cells below **From**. Enter a time between 0:00 o'clock and 24:00 o'clock. You can only enter hours and minutes, not hours, minutes and seconds.
 - To create a shift located within one day, type in a greater number for the time it ends than for the time it starts. You might, for example, define a **Morning** shift that starts at 6:00 o'clock in the morning and ends at 14:00 o'clock on the same day.

- To create a shift that spans two days, type in a smaller number for the time it ends than for the time it starts. You might, for example, define a **Graveyard** shift that starts at 22:00 o'clock at night on one day, and ends at 6:00 o'clock in the morning on the next day.
- Type the times at which the shifts end into the cells below **To**. Enter a time between 0:00 o'clock and 24:00 o'clock. You can only specify hours and minutes, not hours, minutes and seconds.
- To select the individual days on which the shift in the row is active, click in the cells below the days of the week.

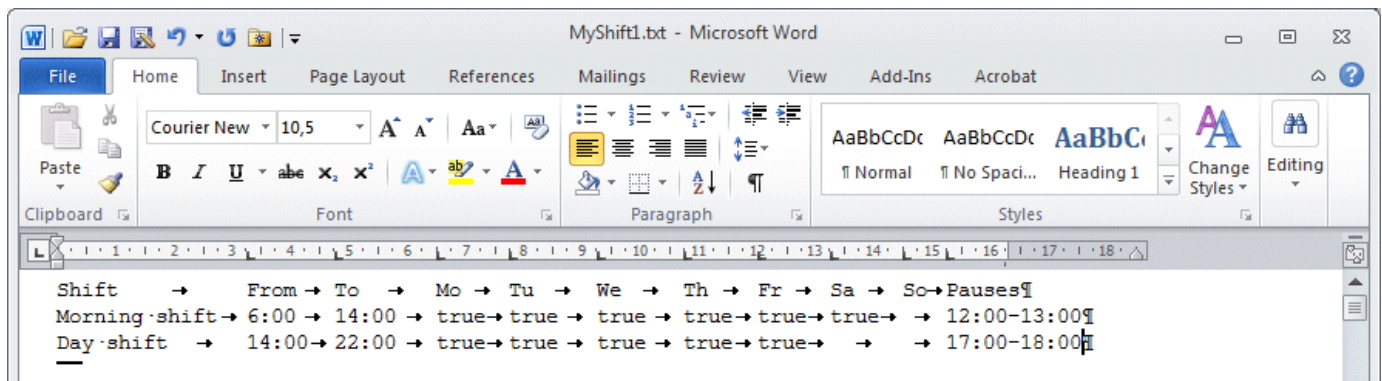
You might, for example:

- Define a **Morning shift** that works from **Monday** up to and including **Saturday**.
- Define an **Evening shift** that works from **Monday** up to and including **Friday**.
- Type the times of the breaks for each shift into the cells below **Pauses**. Enter the hour and the minute when the break starts, a hyphen and the hour and minute the break ends. When you define several breaks during one work day, separate them with a semicolon.

To define a coffee break from 9 o'clock to a quarter past 9 o'clock, and a lunch break from 12 o'clock to a quarter till 1 o'clock, enter 9:00-9:15;12:00-12:45.

To make the *ShiftCalendar* check if the values you entered for the breaks are plausible or not and if you used the correct format to enter the break times, click **Apply**.

You can also import the settings for the individual shifts, which you saved as a tab-delimited text file or which you received from factory management. Type in the data, so that it matches the screenshot below. Or you can export a shift, which you defined, and then import it into another simulation model. To do so, right-click in the list field and select **Export** or **Import** on the context menu.



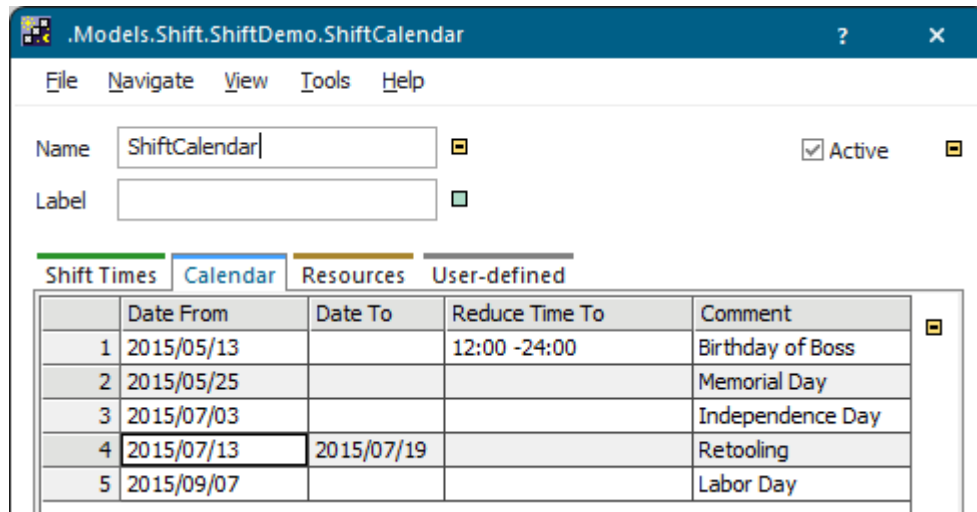
Compare [Context Menu of Embedded Lists](#)

Back to [Define Shifts with the ShiftCalendar](#)

Specify Times During Which the Plant Works Part of the Time

You can specify times during which the plant works part of the time.

Before you can type in data pertaining to public holidays, retooling periods and times during which the plant only works part of the time, click the check box **Inheritance**, so that it looks like this .



	Date From	Date To	Reduce Time To	Comment
1	2015/05/13		12:00 -24:00	Birthday of Boss
2	2015/05/25			Memorial Day
3	2015/07/03			Independence Day
4	2015/07/13	2015/07/19		Retooling
5	2015/09/07			Labor Day

- To type the date when your plant starts not working into the cells below **Date From**:
 - Double-click into the cell and click the down arrow .
 - Select a date in the calendar. Click the right or left buttons to move to another month.

To designate an entire day as a day-off, only type in a start date and no end date (**Date To**), and no **Reduce Time To**.

- To type the date when your plant stops not working into the cells below **Date To**:
 - Double-click into the cell and click the down arrow .
 - Select a date in the calendar. Click the right or left buttons to move to another month.
- To designate a single day on which your plant only works part of the time, type its date in the cell in the column **Date From**. Type the hour and the minute when the reduced working time starts, a hyphen and the hour and minute the reduced time ends into the cells below **Reduce Time To**.

If your plant only works half a day on Christmas Eve, for example the shifts from midnight to noon, you would type 0 : 00 - 12 : 00 into the cell **Reduce Time To**.

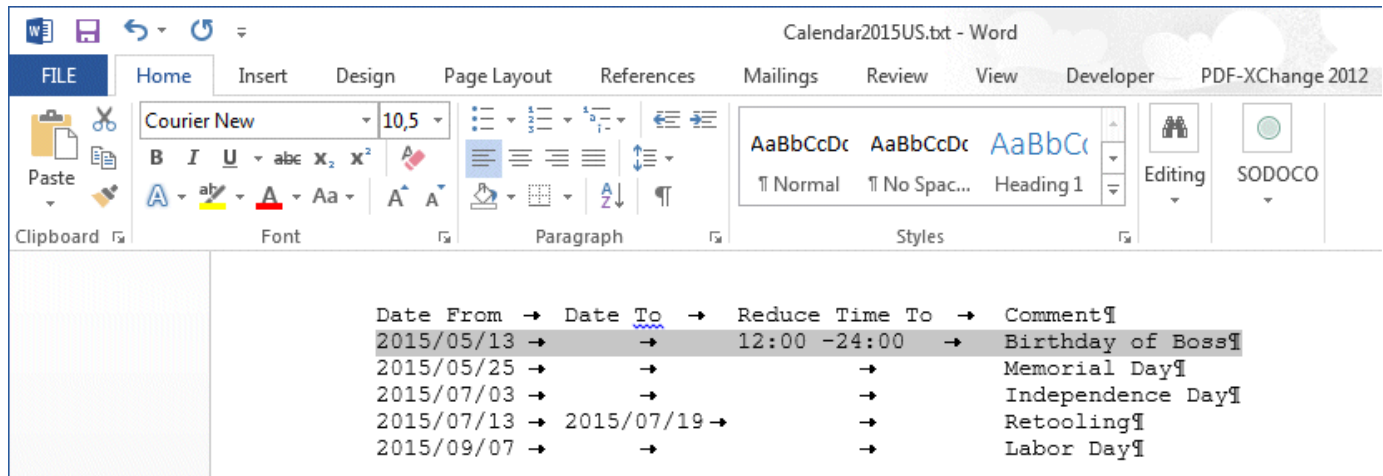
Note

The *ShiftCalendar* combines the reduced time and the definition of the shifts for a day. If, for example, the start time of a day with a reduced working time falls on a break, then this work day starts with a break.

- Type a description of the event of why your plant does not work into the cells below **Comment**.

You can also import a calendar, which you saved as a tab-delimited text file or which you received from factory management. Type in the data, so that it matches the screenshot below. Or you can export a

calendar you defined and then import it into another simulation model. To do so, right-click in the list field and select **Export** or **Import** on the context menu.

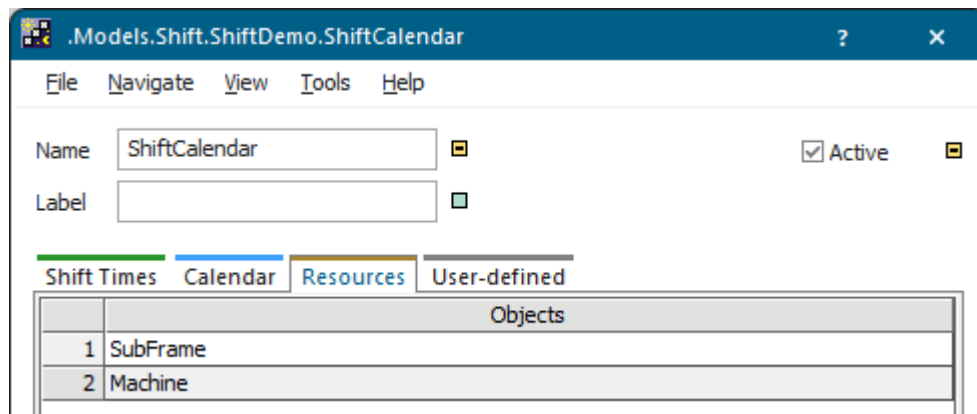


Back to Define Shifts with the ShiftCalendar

Specify the Stations Which the ShiftCalendar Controls

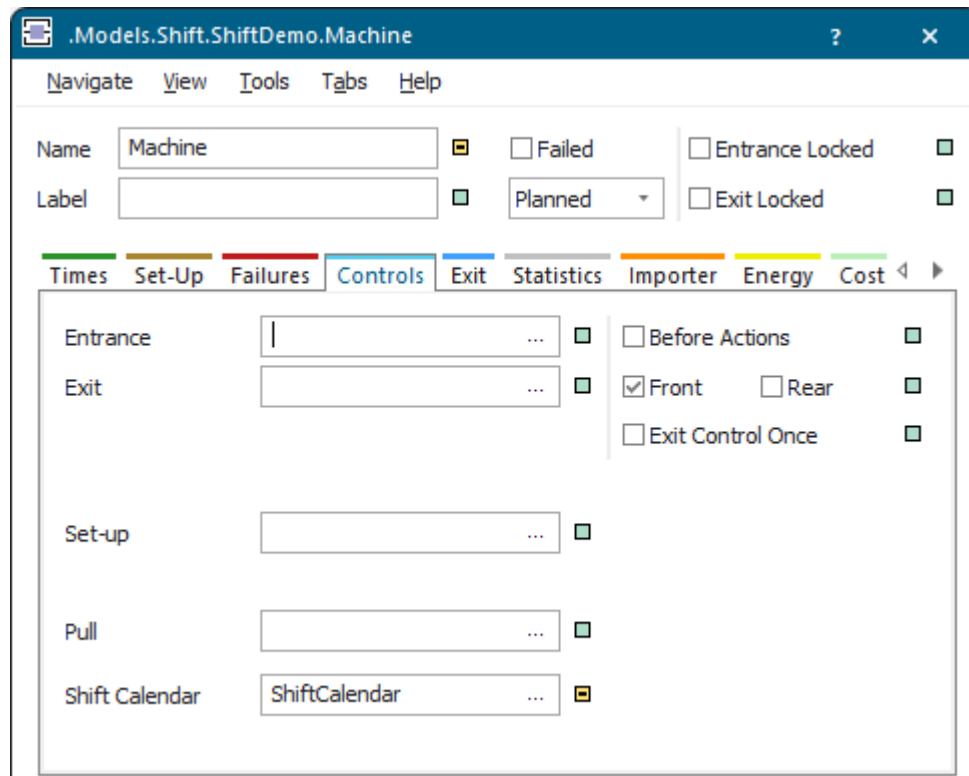
You can specify the stations which the *ShiftCalendar* controls.

Enter the name of any of the built-in material flow objects, or enter the name of a *Frame* you used to model a machine, whose working hours you would like to control with the *ShiftCalendar* into the cells of the list.



This automatically enters the *ShiftCalendar*:

- Into the dialog **Select Shift Calendar** that the command **Select Shift Calendar** on the **Frame** ribbon tab of the *Frame* opens.
- Into the text box **Shift calendar** on the tab **Controls** of the material flow object.



Back to [Define Shifts with the ShiftCalendar](#)

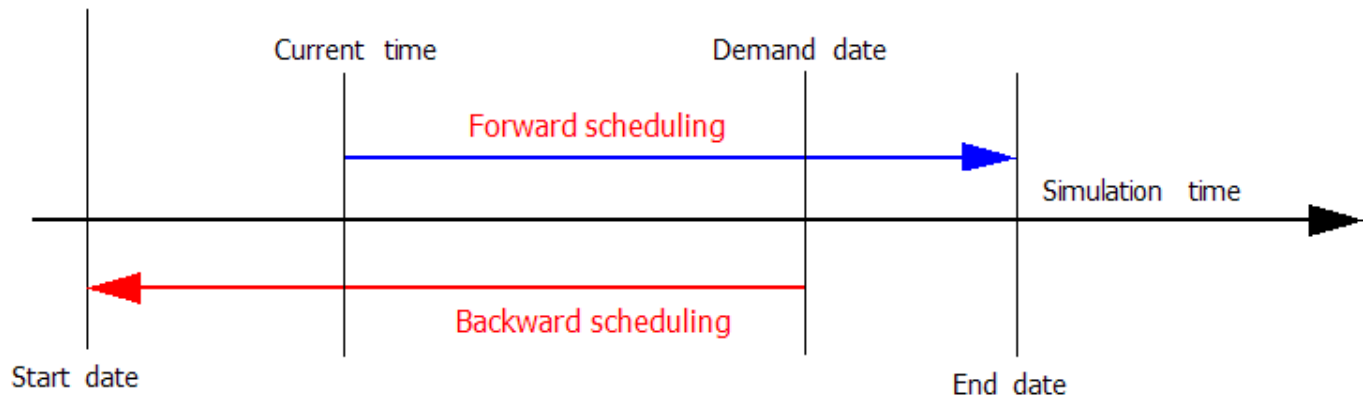
Schedule Date and Time to Start or Finish the Production Process

You can schedule date and time to start or finish the production process with the method *schedule*.

Use the method [schedule](#) to set the date and the time to start or to finish the production process. We distinguish between:

- **Forward scheduling**, beginning from the start date forward into the future, and
- **Backward scheduling**, which computes the start date going backward in time from the demand date.

Normally you start with the demand date, and compute the start date through backward scheduling. If this start date is located in the past though, you have to recompute the dates beginning with the present time and forward-schedule the end date.



In our example below we programmed the method *schedule* to compute the end date of two jobs taking into account the shifts we defined in the *ShiftCalendar*.

The source code looks like this:

```
var startTime := str_to_dateTime( "4.1.2013 0:00" )
var durationTime := str_to_time( "10:00:00.0" )

var EndTime := ShiftCalendar.schedule( startTime, durationTime, "forward" )
print "StartDate: ", startTime, "    Duration: ", durationTime, "    EndDate: ", EndTime

startTime := str_to_dateTime( "20.12.2013 0:00" )
durationTime := str_to_time( "19:00:00.0" )

EndTime := ShiftCalendar.schedule( startTime, durationTime, "forward" )
print "StartDate: ", startTime, "    Duration: ", durationTime, "    EndDate: ", EndTime
```

The first job starts on January 4, 2013 at midnight and takes 10 hours to finish. The second job starts on December 12, 2013 and takes 19 hours. We want to know on what days and at which times the parts will be all processed, considering the morning and day shifts we defined in the *ShiftCalendar*.

We also programmed the *Method* to print the results of its computations to the *Console*.

Console

```
StartDate: 2013/01/04 00:00:00.0000 Duration: 10:00:00.0000 EndDate: 2013/01/04 17:00:00.0000
StartDate: 2013/12/20 00:00:00.0000 Duration: 19:00:00.0000 EndDate: 2013/12/23 11:15:00.0000
```

Back to [Define Shifts with the ShiftCalendar](#)

Pausing Material Flow Objects and Frames

Pause Material Flow Objects and Frames

The *ShiftCalendar* interacts with the material flow objects and the *Frames* it controls by changing their **Paused/Planned/Unplanned** state.

One of these objects is:

- **Paused** if it is not processing parts due to a pause. It resumes processing once you terminate the pause, either by selecting **Planned** from the drop-down list or by setting its attribute **Pause** to `false`.

Note

When a station fails, and a pause occurs during the failure, the failure time of the station will still be consumed, although the station is paused. Statistics counts the times during which these two states overlap as **paused** time. **Resetting** your model removes both failures and pauses of the object.

- **Unplanned** if it is not scheduled to work during the shifts which you defined in the *ShiftCalendar*.
- **Planned** if it is scheduled to work during the shifts which you defined in the *ShiftCalendar*. The scheduled/planned time is the processing time minus the times of the breaks.

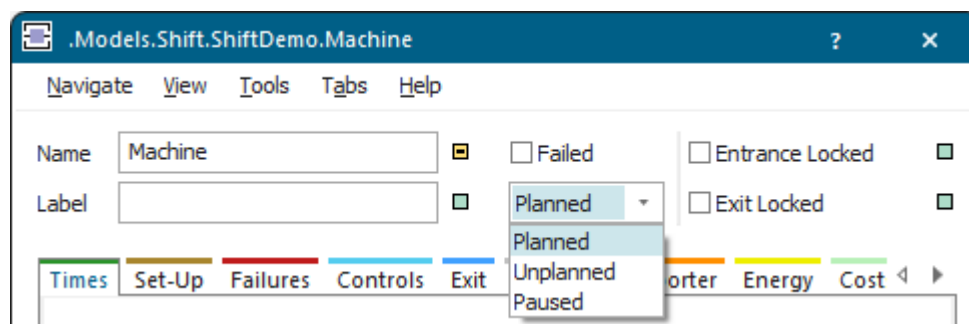
We distinguish between:

- **Paused Material Flow Objects** and
- **Paused Frames**

Back to [Model a Shift System](#)

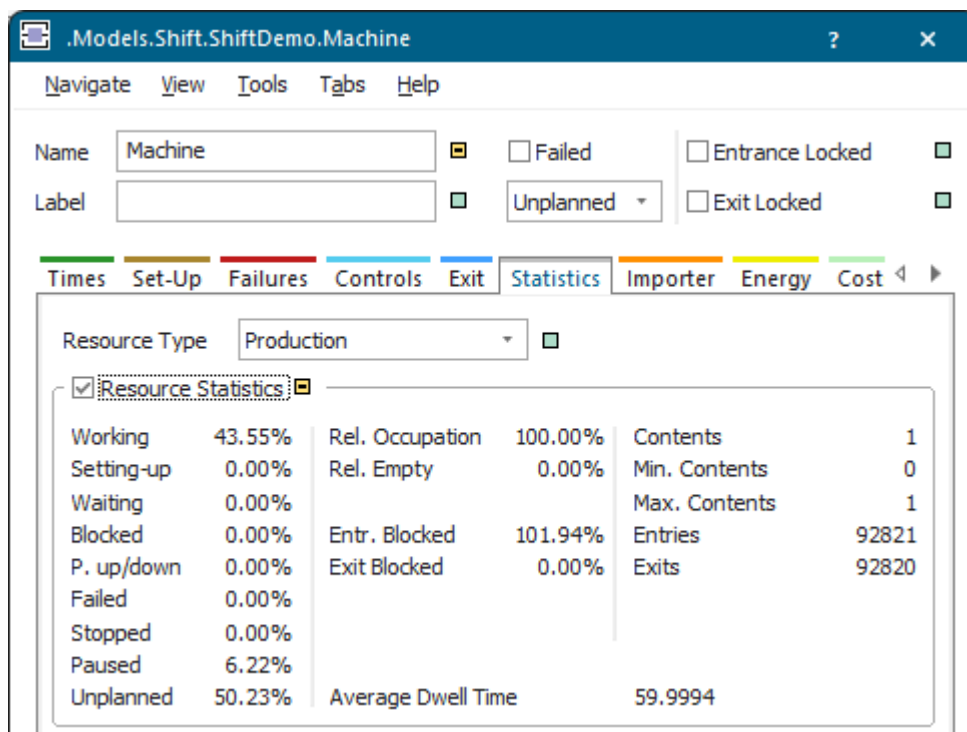
Paused Material Flow Objects

The *ShiftCalendar* interacts with the material flow objects by changing their **paused** state according to the shifts you defined. During a simulation run the dialog of these objects also reflects this by selecting the present state in the drop-down list.



When it is paused, the material flow object does not receive any mobile parts. *MUs* can exit the object, when you program a *Method* to make them do so. *Plant Simulation* stops set-up and processing until the end of the pause or of the unplanned time.

The unplanned state is identical to the paused state. The only difference is how *Plant Simulation* counts the internal statistics of the objects. When you define shifts that are active from 6 o'clock in the morning until 10 o'clock at night, for example, the planned time lasts from 6 o'clock to 22 o'clock. The unplanned time lasts from 22 o'clock to 6 o'clock the following morning.



When it is planned/scheduled to work, the object receives and processes *MUs* and moves them on to the succeeding object in the material flow.

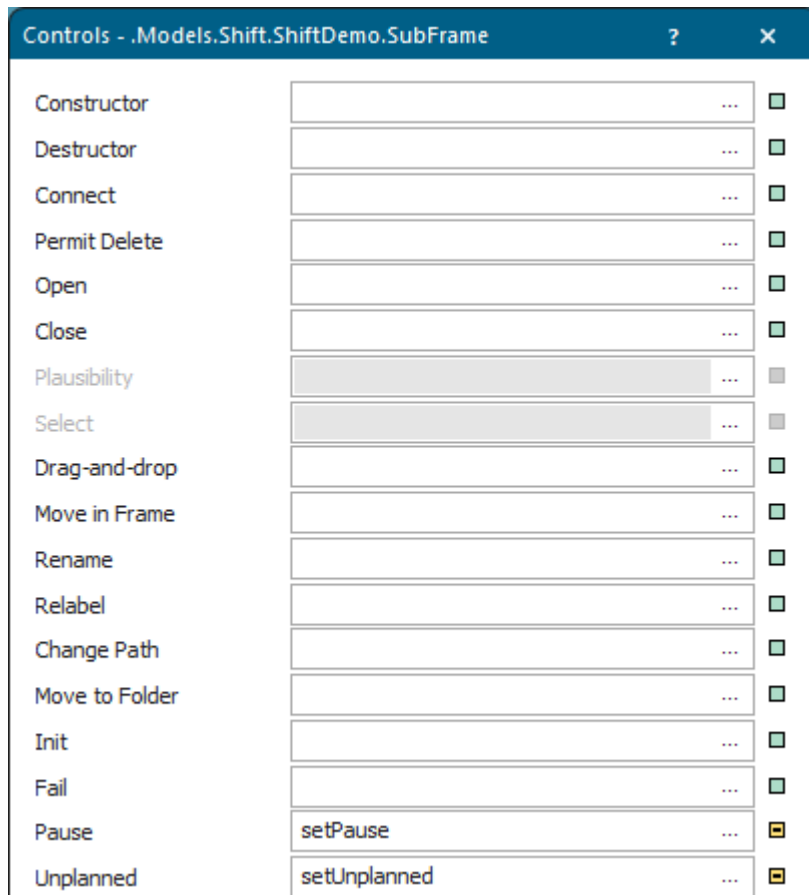
Back to [Pause Material Flow Objects and Frames](#)

Paused Frames

You can pause a *Frame* with its attributes *Pause* and *Unplanned*.

As opposed to the material flow objects, which you can also pause manually, you can only change the **paused** and the **unplanned states** of the *Frame* with its attributes **Pause** and **Unplanned**.

In addition, you can program a pause and/or an unplanned time control, both of which activate a *Method* in changing the value of the *attribute*.



Example of a Pause Control

You can change the paused state of the *Frame* by assigning the value `true` or `false` to the attribute **Pause**. Use the anonymous identifier `?` to access the object from within a control.

Plant Simulation executes the **Pause Control**, whenever the *paused* state of the *Frame* changes. When you get the paused state within the *Method*, it shows the state after the change.

The source code looks like this:

```
print "Current pause ", current.pause
MyStation.pause := current.pause
var shift := root.ShiftCalendar.getCurrShift
print "Current shift: ", shift

if not current.unplanned
  if current.pause
    current.currIcon := "pause"
  else
    current.currIcon := "working"
  end
end
end
```

Example of an Unplanned Control

Plant Simulation executes the **Unplanned Control**, whenever the *unplanned* state of the *Frame* changes, i.e., whether the *Frame* is not scheduled to work during the time you entered or whether it is scheduled to work.

You can change the unplanned state by assigning the value `true` or `false` to the attribute **Unplanned**. Use the anonymous identifier `?` to access the object from within a control.

The source code looks like this:

```
print "Frame unplanned: ", current.unplanned
MyStation.unplanned := current.unplanned
if current.unplanned
    current.currIcon := "unplanned"
else
    if current.pause
        current.currIcon := "pause"
    else
        current.currIcon := "working"
    end
end
```

Back to [Pause Material Flow Objects and Frames](#)

Modeling a Lockout Zone

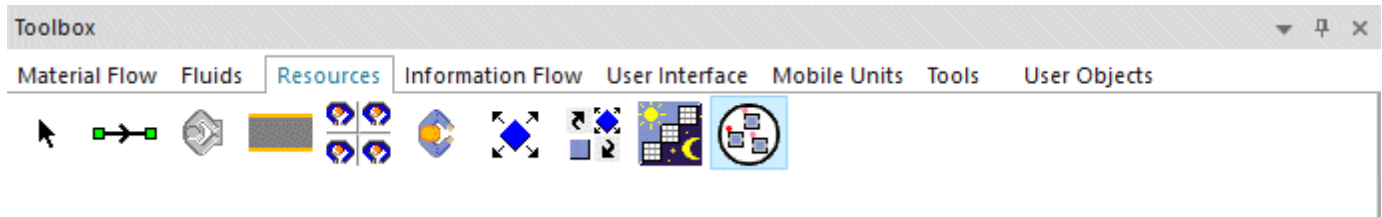
Model a Lockout Zone

The *LockoutZone* combines a group of material flow objects. If one of these stations fails, all other stations that are part of the lockout zone stop processing their parts.

You have to define *failure profiles* for at least one of the stations assigned to the *LockoutZone*. As soon as one of the stations fails, the *LockoutZone* stops all processing operations of all assigned stations. You can select if it stops the processing operations immediately or when the required service arrives.

The stations only start processing parts again after all failures were removed. They then only use up the respective remaining processing time. You can assign any of the built-in **Material Flow Objects** or a station you modeled in a *Frame* to the *LockoutZone*.

You can insert the **LockoutZone**  into your simulation model from the folder **Resources** in the *Class Library* or from the toolbar **Resources** in the *Toolbox*.



You can:

- [Specify the Stations Which the LockoutZone Stops](#)
- [Create a Failure Profile for One of the Stations](#)
- [Stop the Associated Stations Immediately After a Failure](#)
- [Stop the Associated Stations When the Repair Man Arrives](#)
- [Use a Stop Processing Control](#)
- [Use a Resume Processing Control](#)

Go to [Stopped Material Flow Objects and Frames](#)

Back to [Modeling the Material Flow](#)

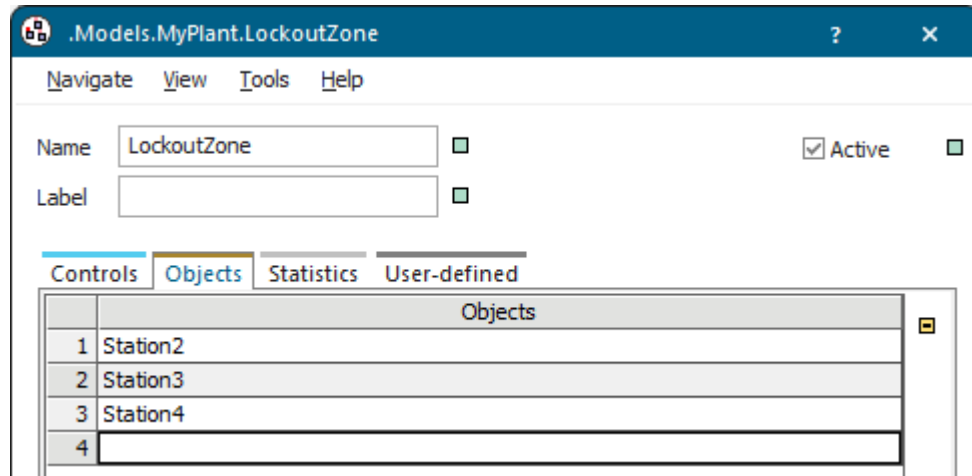
Specify the Stations Which the LockoutZone Stops

In our sample model the processing stations **Station2**, **Station3**, and **Station4** are part of the lockout zone. If **Station3** fails, the lockout zone stops the processing operations of **Station2** and **Station4**.

The percentages of the stopped time of **Station2** and **Station4** have to match the percentage of the **Failed** time of **Station3**. The total availability of the stations within the lockout zone matches the availability of **Station3**.

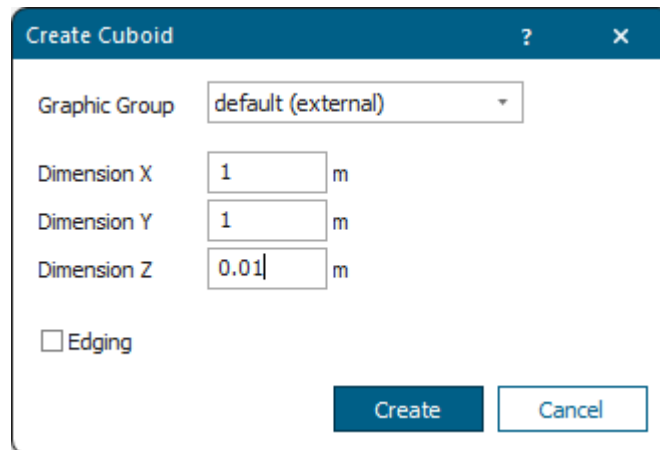
To add a station as a resource to the lockout zone:

- Drag the icon of that object, in our case **Station2**, over the icon of the *LockoutZone* and drop it there.
- Repeat this with **Station3** and **Station4**.

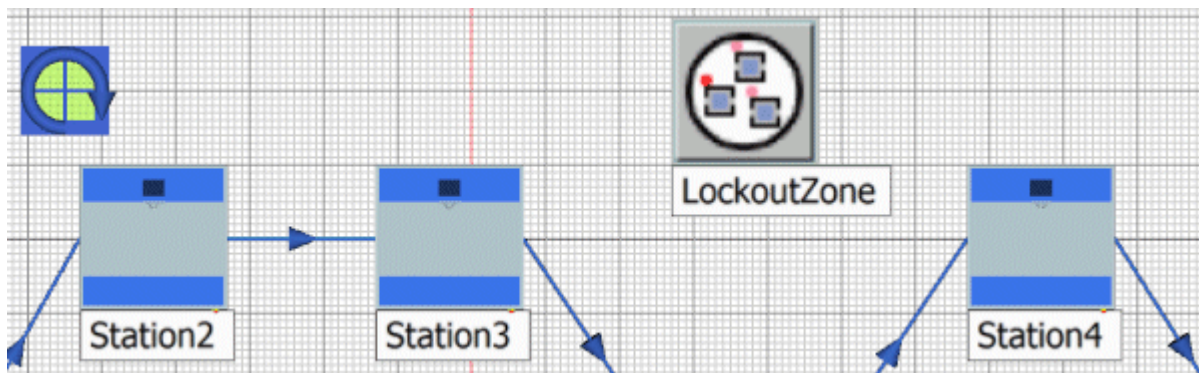


To clarify which objects are part of the lockout zone, you might, for example, insert a **Cuboid** into the *Frame* and assign a **Material** to it. Proceed as follows:

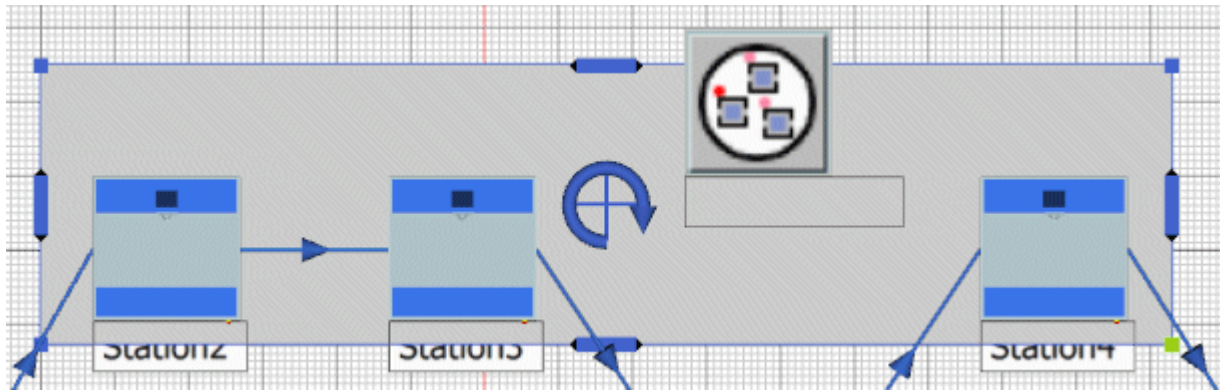
- Click the **Cuboid** on the ribbon tab **Edit**. Type in a small value for the **Dimension Z**, the height of the **Cuboid**.



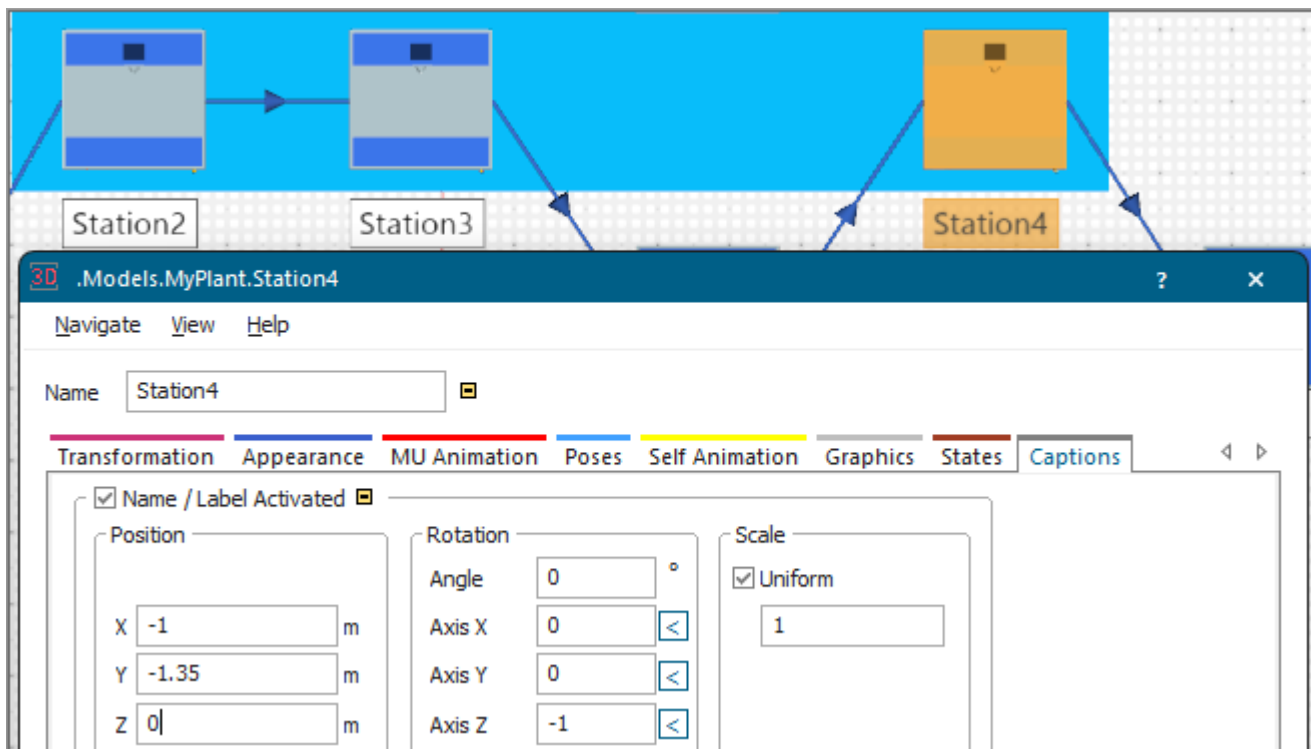
- Insert the **Cuboid** above *Station2* and press the **M** key to show the manipulators.



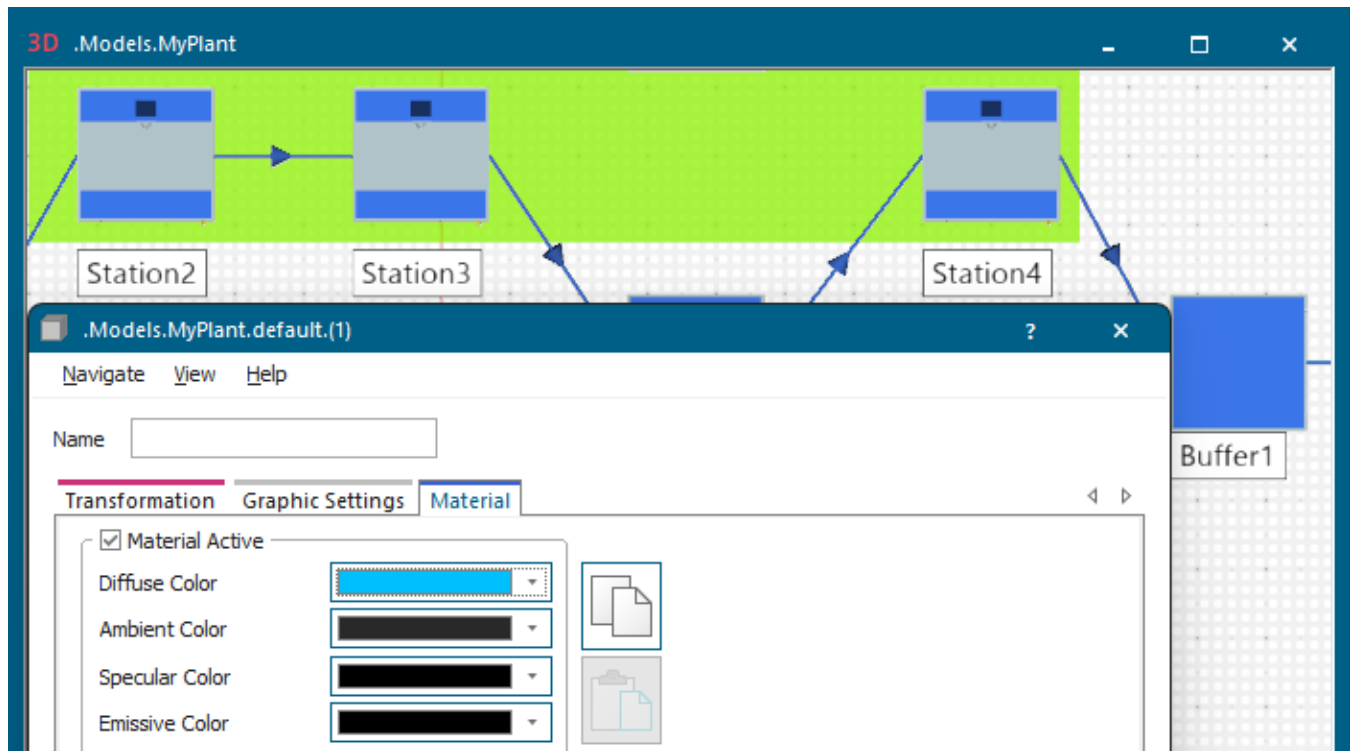
- Click the bottom right corner of the **Cuboid** and drag it, so that it covers the *LockoutZone* and the *Stations*.



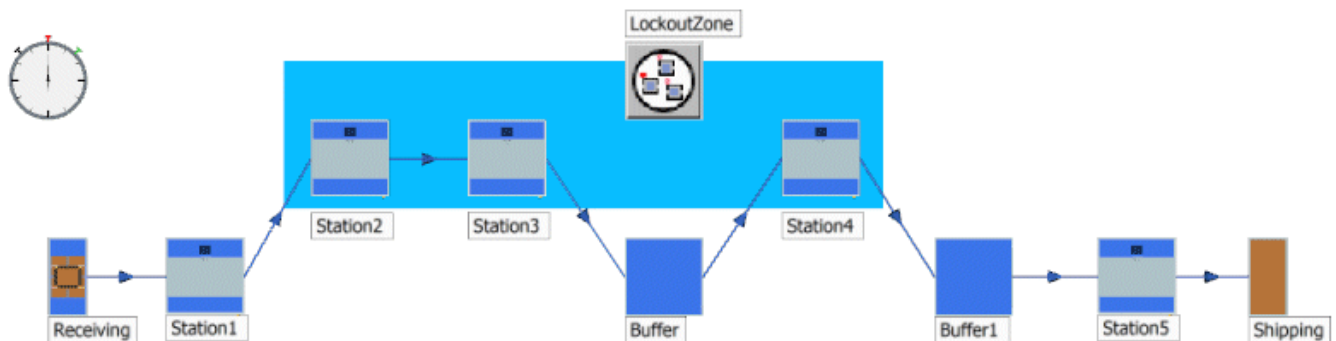
- As our **Cuboid** overlaps the captions of the objects, we move it, so that they become visible. Select *Station1*, press the **spacebar** and change to the tab **Captions** in the dialog **Edit 3D Properties**. We selected **-1.35** as the **Y position** for the three affected *Stations*.



- We finally have to adjust the **position** of the **caption** of the *LockoutZone* and assign a **Material/color** to the **Cuboid**.



- The result looks like this:



Back to [Model a Lockout Zone](#)

Create a Failure Profile for One of the Stations

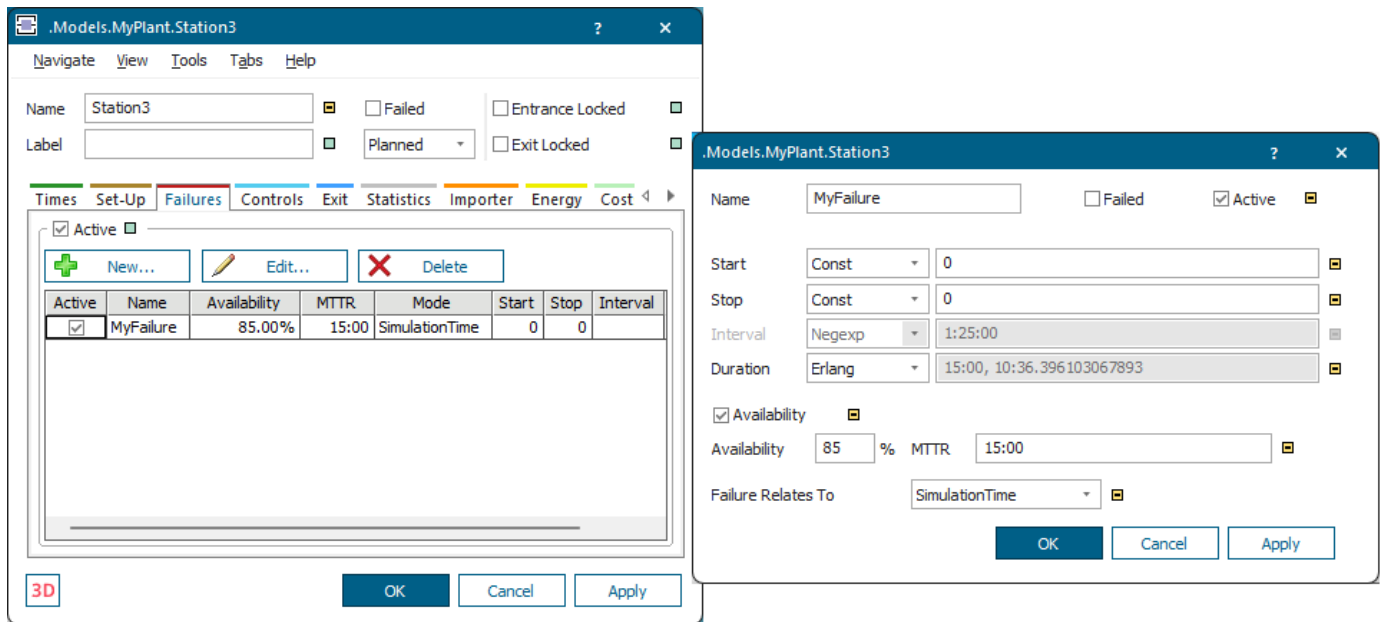
To make the *LockoutZone* stop the processing operations of the stations it controls, we have to create a *failure profile* for one of the stations.

To make one of the stations, in our example **Station3**, fail, proceed as follows:

- Click the tab **Failures** and click **New**.
- Enter an **availability** and a **mean time to repair**. We entered an availability of 85 percent and a mean time to repair of 15 minutes.

Stop the Associated Stations Immediately After a Failure

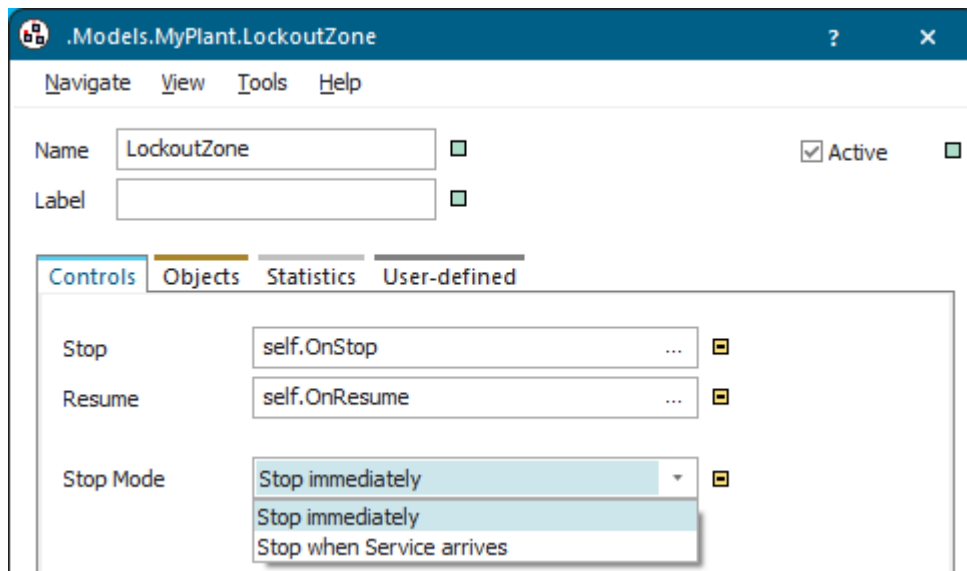
- Click **OK** to close the dialog.



Back to [Model a Lockout Zone](#)

Stop the Associated Stations Immediately After a Failure

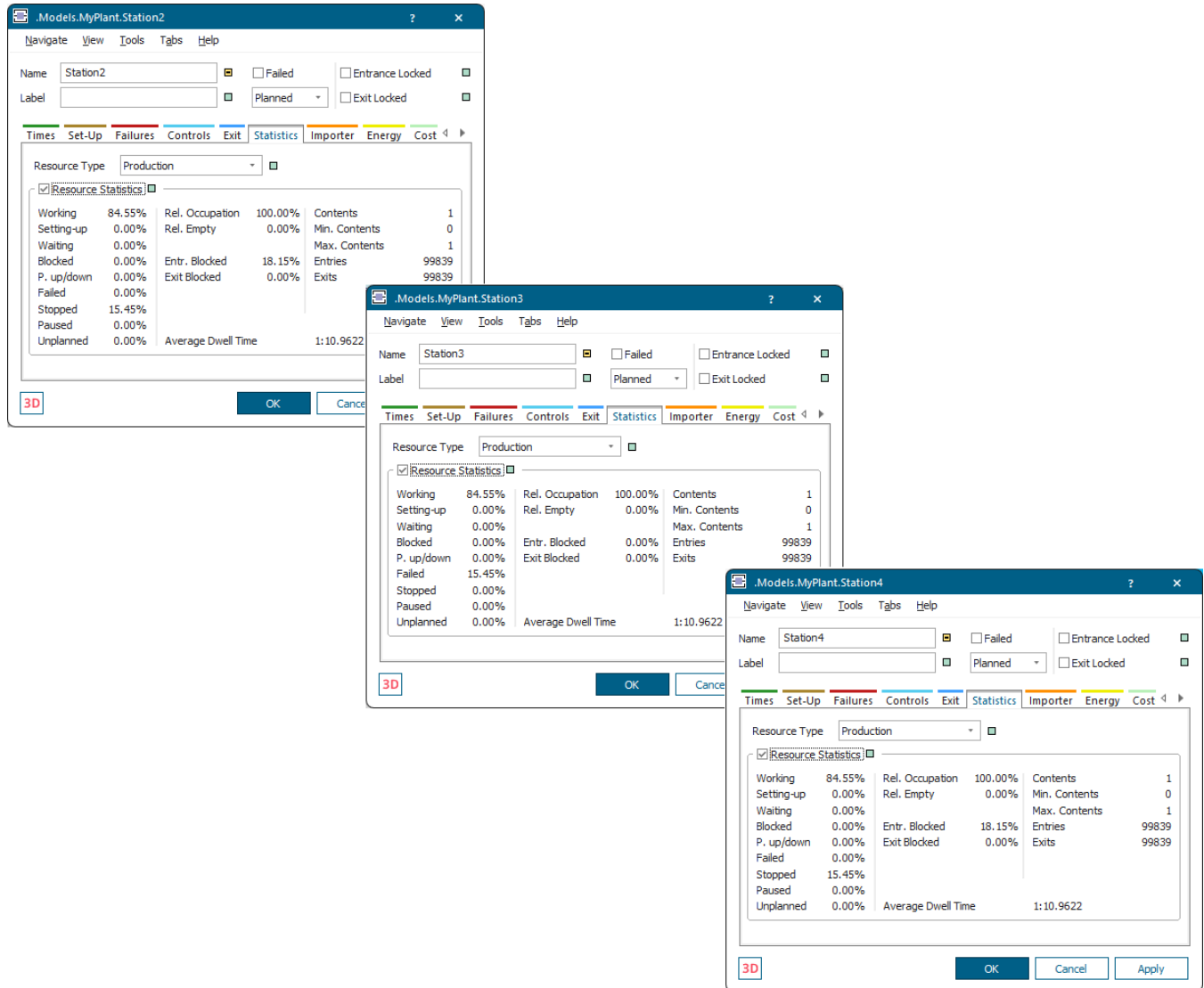
To make the *LockoutZone* stop the processing operations of the stations it controls immediately, select the **Stop mode > Stop immediately**.



The *LockoutZone* then immediately stops the processing operations of **Station2** and **Station4** as soon as **Station3** fails. *Plant Simulation* only stops the stations that are part of the *LockoutZone*, but none of the other stations in the model!

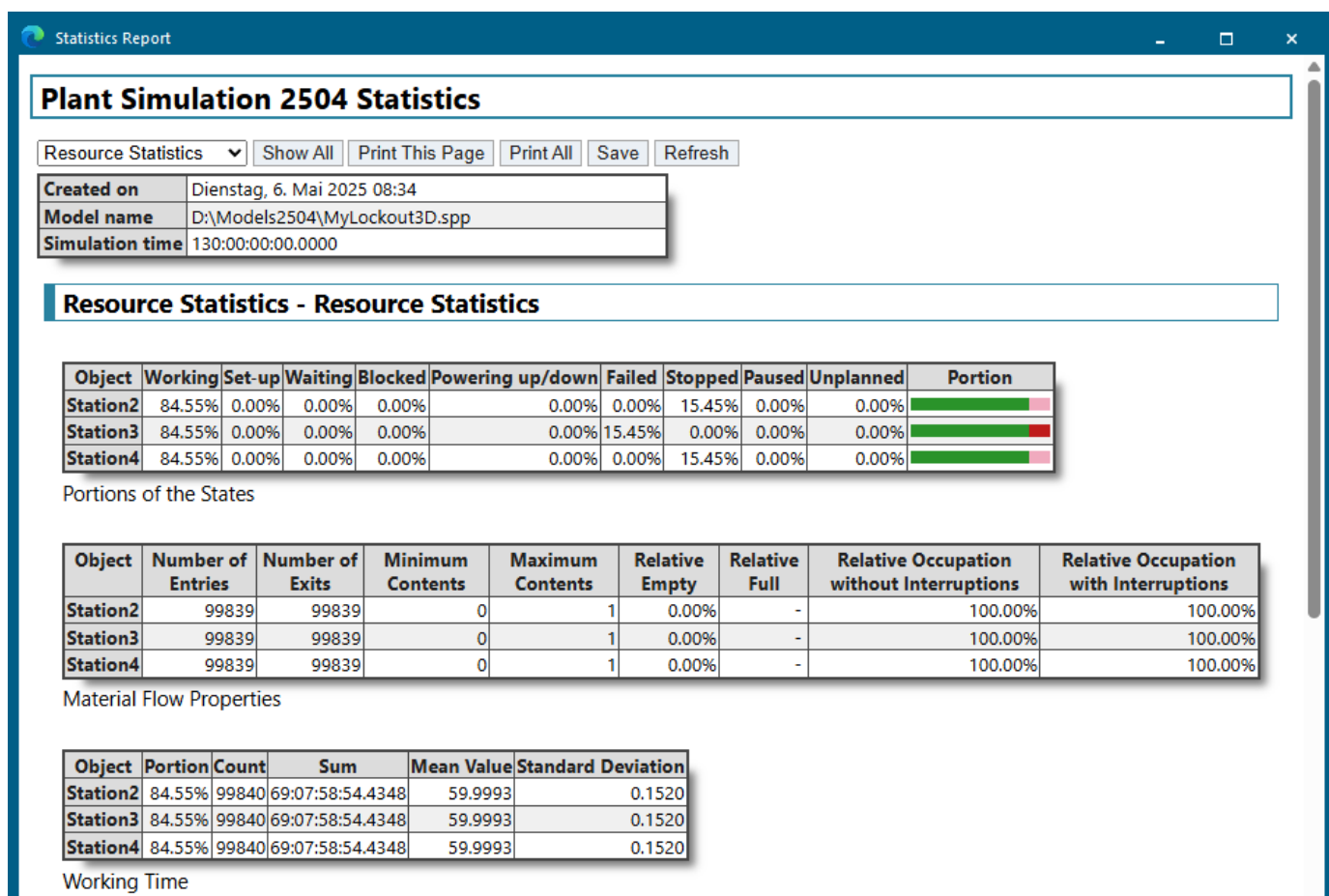
Then, check the statistics values of the stations, which you assigned to the *LockoutZone*:

- Open the dialogs of **Station2**, **Station3**, and **Station4** and change to the tab **Statistics**.
- You'll notice that the **stopped** percentages of **Station2** and **Station4** match the **failed** percentage of **Station3**. This means that the overall availability of **Station2**, **Station3**, and **Station4** matches the availability of the failed station, namely **Station3**.



The *statistics report* show additional statistics data about the **stopped** and **failed** times.

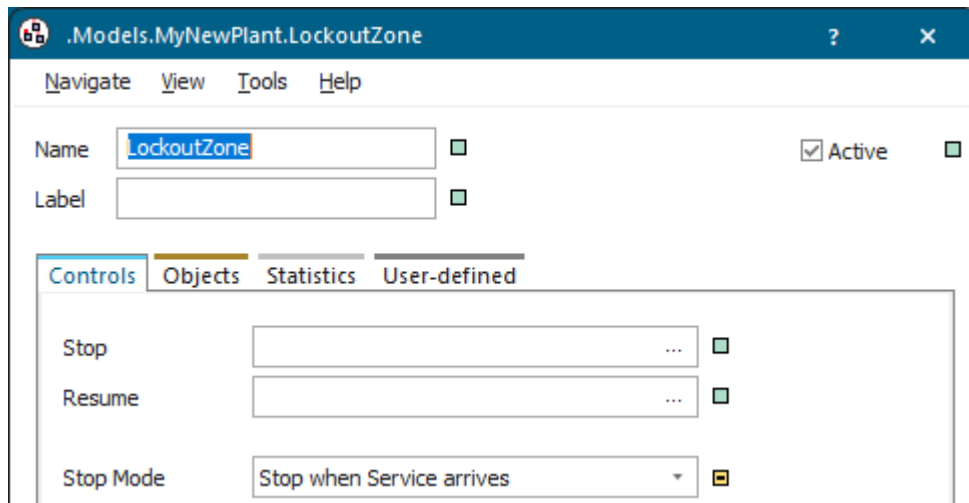
To open it, select the stations in the *Frame* and press **F6**.



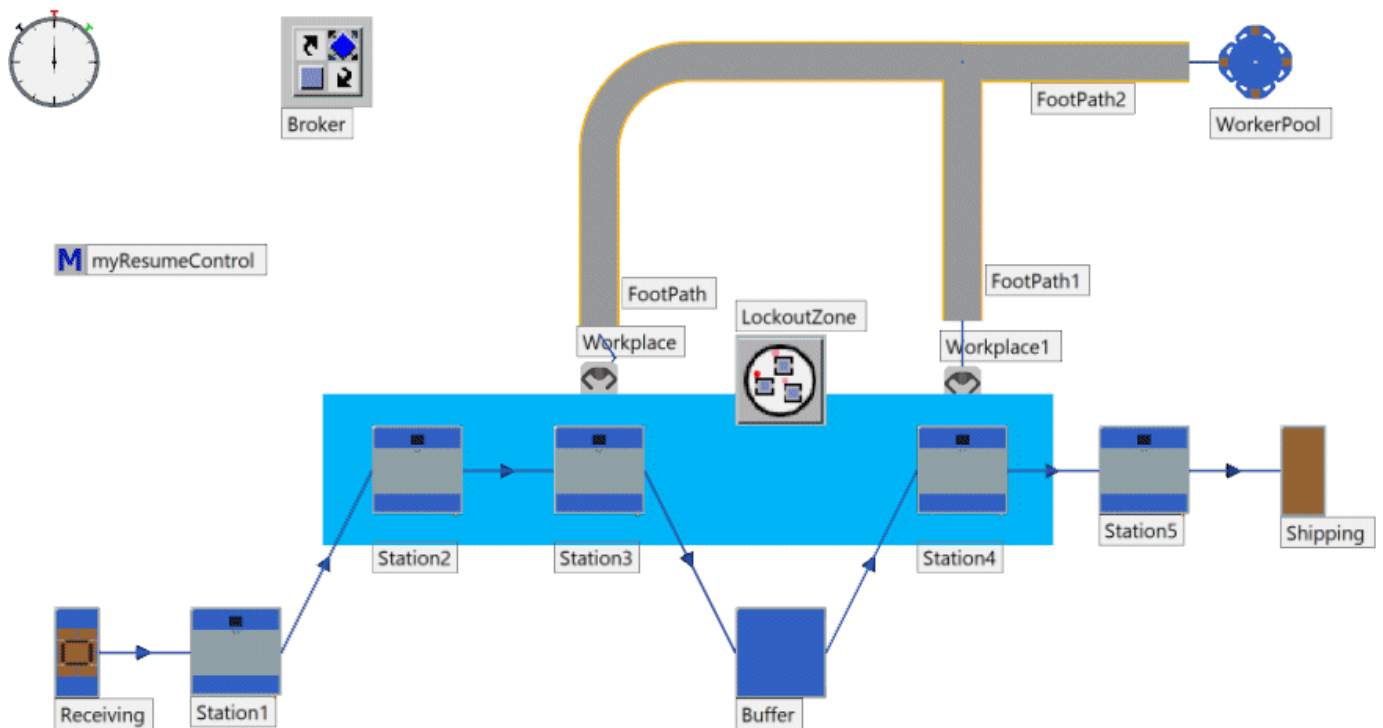
Back to [Model a Lockout Zone](#)

Stop the Associated Stations When the Repair Man Arrives

To make the *LockoutZone* stop the processing operations of the stations it controls once the service technician, who repairs the failed stations, arrives, select the **Stop mode > Stop when Service arrives**.



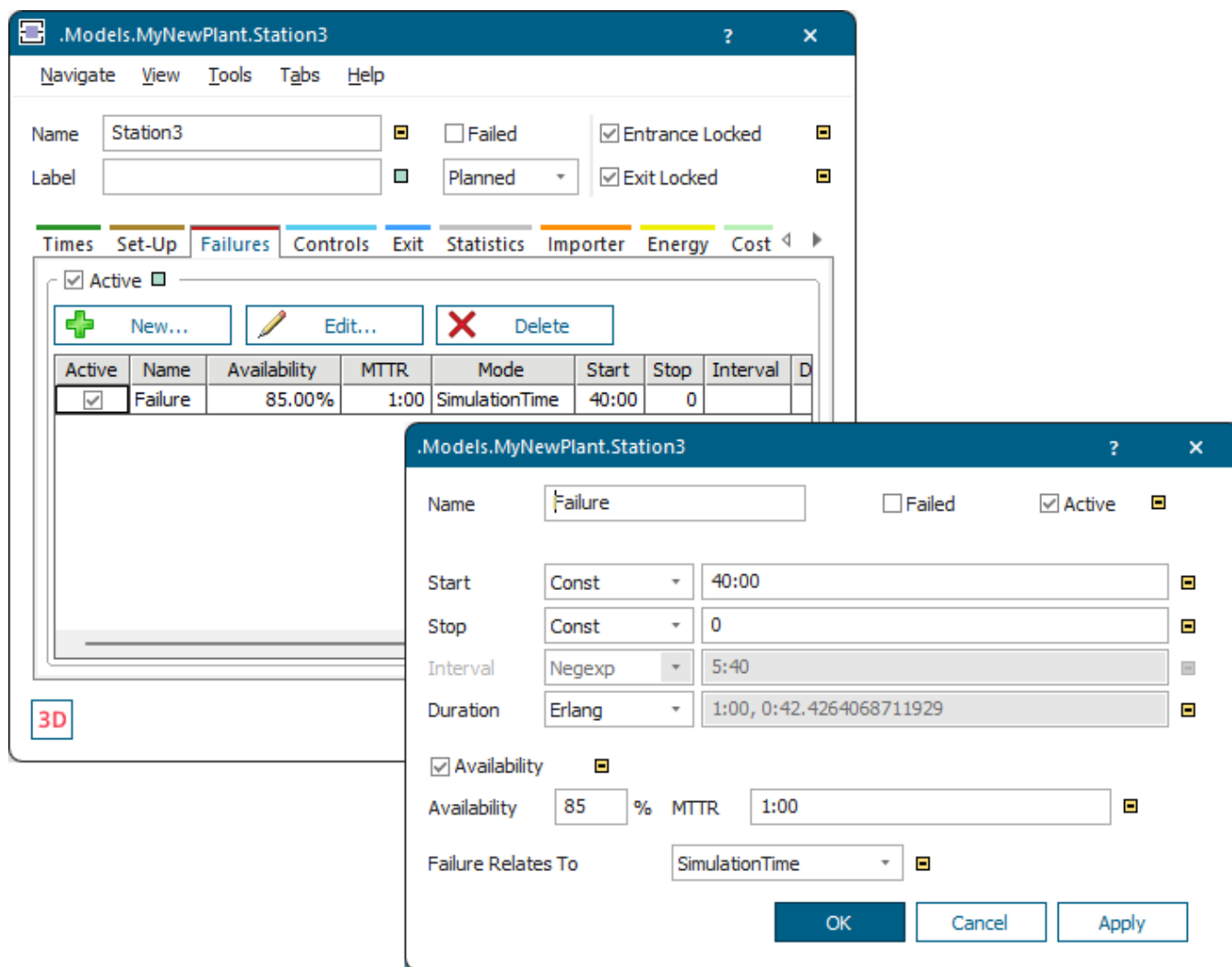
In our example the *LockoutZone* stops the processing operations of the assigned stations, namely **Station2**, **Station3**, and **Station4**, when the repair service, which the failed **Station3** requested, is assigned.



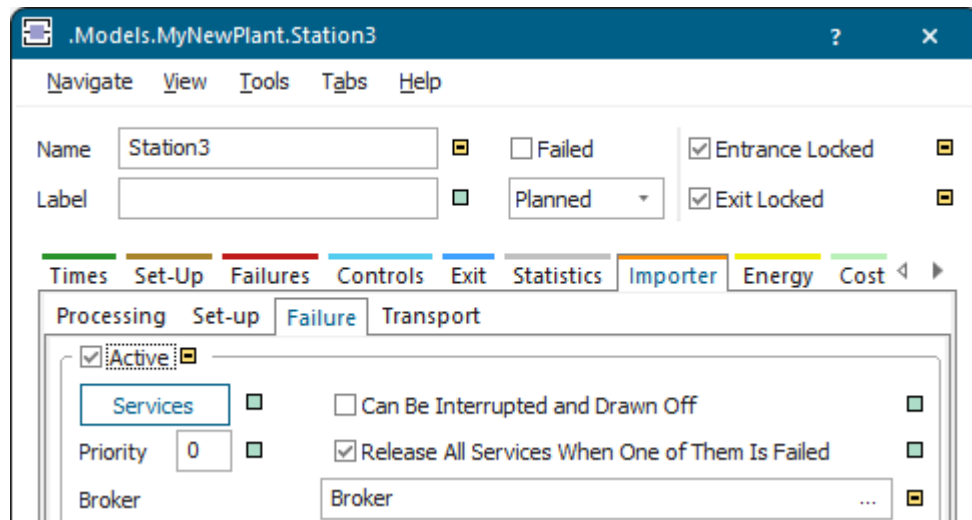
Proceed as follows to create our simple sample model:

- Insert the stations you want to simulate and drag these stations over the icon of the *LockoutZone* to assign them. In our case these are **Station2**, **Station3**, and **Station4**.
- As a service technician is to repair the failed machines, add a *Workplace* to them and connect the *Workplace* with the *WorkerPool* with a *footpath*.

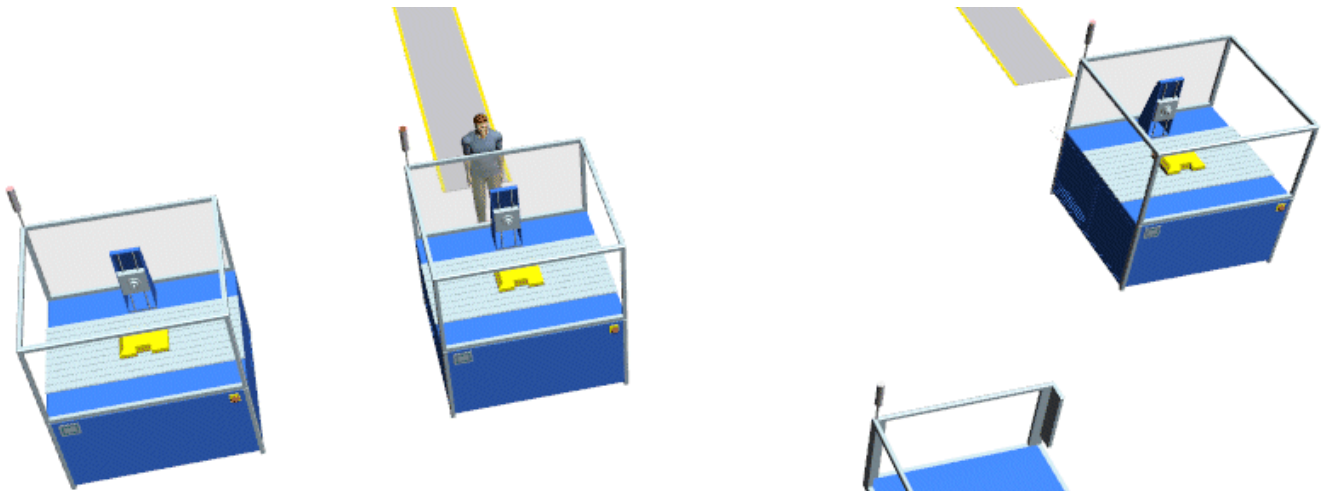
- As we want **Station3** to fail, define a *failure profile* for it.



- To make a service technician repair **Station3** once it fails, insert a *Broker* into the model and assign it on the sub-tab **Failure**. Then **activate** the *failure importer*. There is no need to change any other settings.



- Then, run the simulation by clicking  on the **Home** ribbon tab. Once **Station3** fails, the *Worker* walks from the *WorkerPool* to the failed station on the *footpath*. Once our *service technician* has reached the *Workplace* attached to the failed station **Station2**, the *LockoutZone* stops the associated stations, namely **Station2** and **Station4**.



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Use a Stop Processing Control

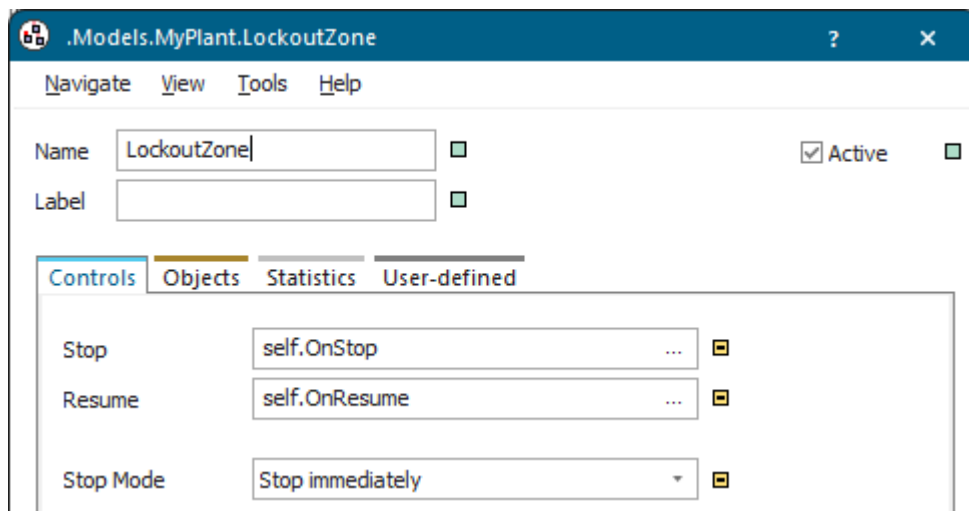
If you want to use your own logic when one of the stations assigned to the *LockoutZone* fails, you have to program a **Stop Control**.

In our example we write the following information into the table **ReportStopping**.

.Models.MyPlant.ReportStopping

	string 1	time 2	string 3	time 4	time 5
string	First Failed Station	Start Stopping	Last Failed Station	End Stopping	Duration
1					
2					
3					

Within the **Stop Control** the anonymous identifier @ designates the triggering station, while the anonymous identifier ? designates the *LockoutZone*.



.Models.MyPlant.LockoutZone ? x

Navigate View Tools Help

Name: LockoutZone ☐ ☒ Active ☐

Label: ☐

Controls Objects Statistics User-defined

Stop: self.OnStop ... ☐

Resume: self.OnResume ... ☐

Stop Mode: Stop immediately ☐

The source code looks like this:

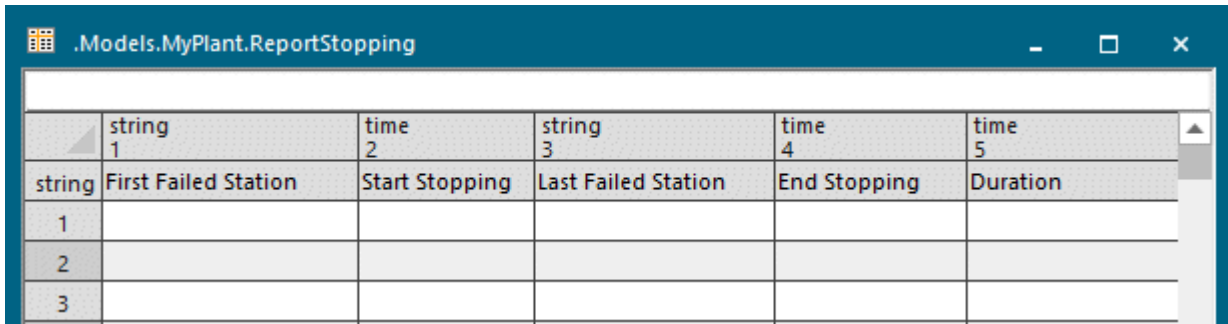
```
var row:integer := ReportStopping.ydim + 1
ReportStopping["First Failed Station",row] := @.name
ReportStopping["Start Stopping",row] := eventcontroller.simTime
```

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Use a Resume Processing Control

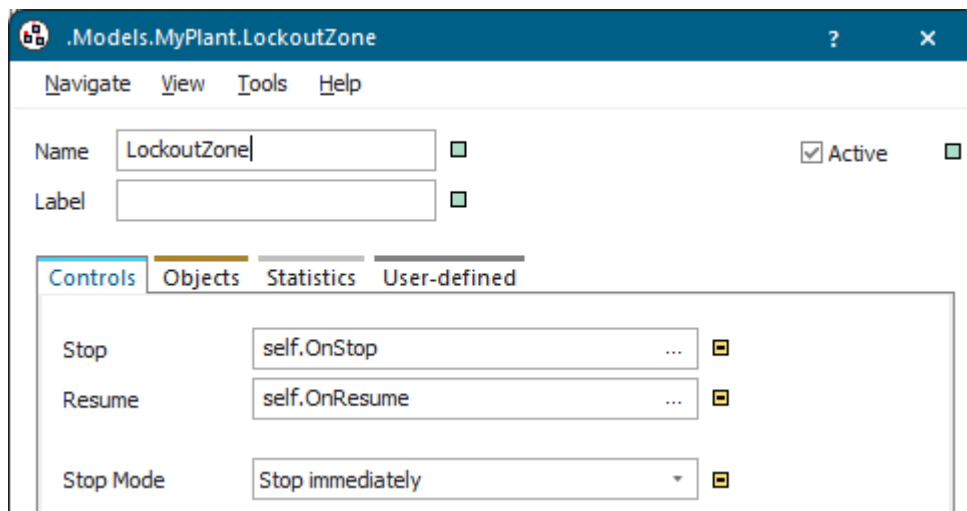
If you want to use your own logic when all failures of all assigned stations were repaired and the stations can thus continue processing parts, you have to program a **Resume Control**.

In our example we write the following information into a table.



	string 1	time 2	string 3	time 4	time 5
string	First Failed Station	Start Stopping	Last Failed Station	End Stopping	Duration
1					
2					
3					

Within the **Resume Control** the anonymous identifier @ designates the triggering station, while the anonymous identifier ? designates the *LockoutZone*.



.Models.MyPlant.LockoutZone ? x

Navigate View Tools Help

Name: LockoutZone ☐ ☒ Active ☐

Label: ☐

Controls Objects Statistics User-defined

Stop: self.OnStop ... ☐

Resume: self.OnResume ... ☐

Stop Mode: Stop immediately ☐

The source code looks like this:

```
var row: integer := ReportStopping.ydim
ReportStopping["Last Failed Station",row] := @.name
ReportStopping["End Stopping",row] := eventcontroller.simTime
ReportStopping["Duration",row] := ReportStopping["End Stopping",row] -
ReportStopping["Start Stopping",row]
```

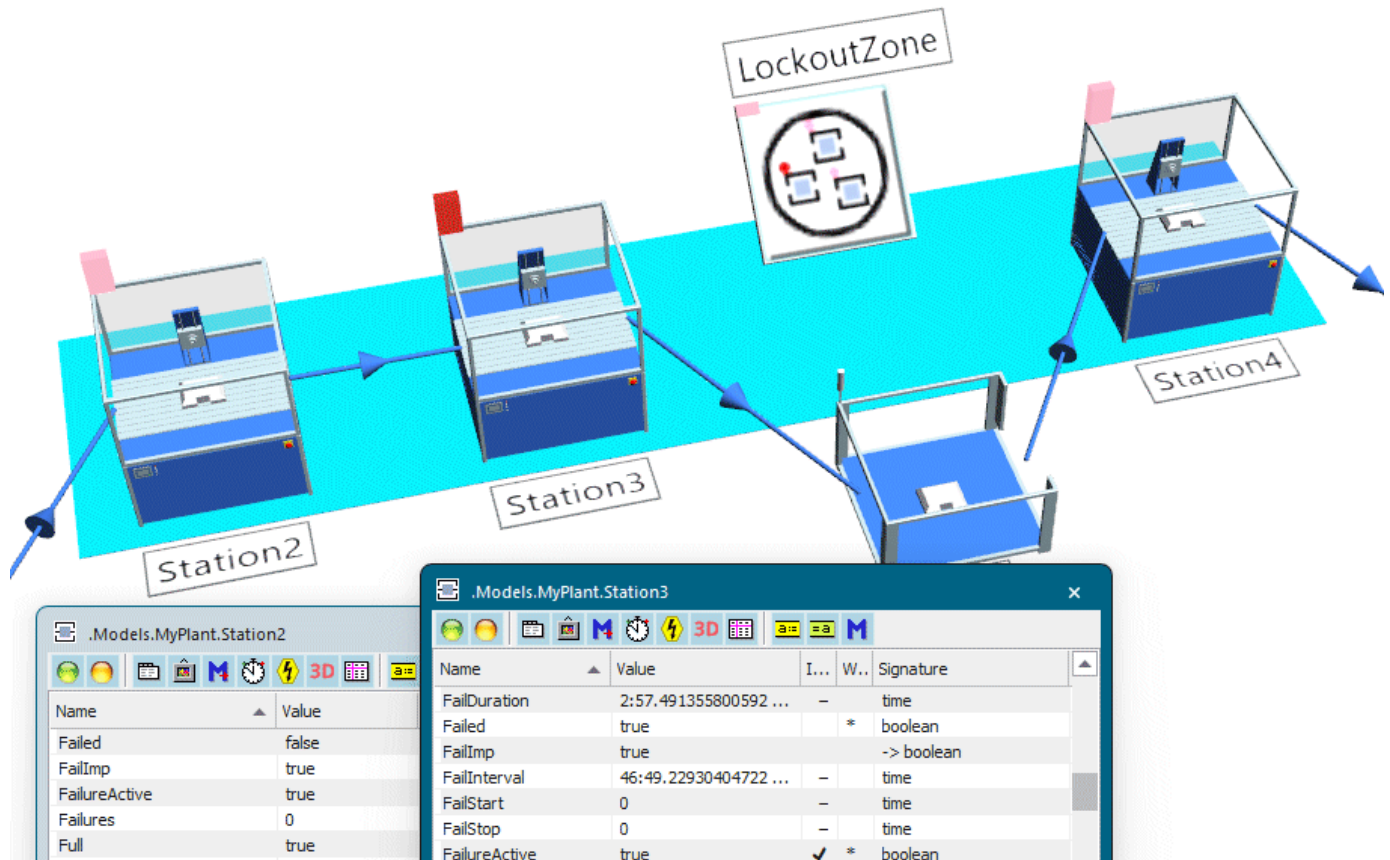
The result of our control looks like this:

	string 1	time 2	string 3	time 4	time 5
string	First Failed Station	Start Stopping	Last Failed Station	End Stopping	Duration
1	Station3	46:49.2293	Station3	49:46.7207	2:57.4914
2	Station3	2:37:45.3691	Station3	2:51:01.2005	13:15.8314
3	Station3	4:15:15.6570	Station3	4:26:19.3290	11:03.6720
4	Station3	5:09:06.6667	Station3	5:22:56.4505	13:49.7837
5	Station3	7:03:53.7183	Station3	7:07:34.3139	3:40.5957
6	Station3	7:37:20.8024	Station3	7:52:08.9484	14:48.1460
7	Station3	12:15:33.0387	Station3	12:30:16.6560	14:43.6173
8	Station3	12:59:54.8513	Station3	13:33:10.8494	33:15.9981
9	Station3	13:51:13.3905	Station3	14:16:51.8352	25:38.4447
10	Station3	17:08:05.0728	Station3	18:16:28.7860	1:08:23.7132

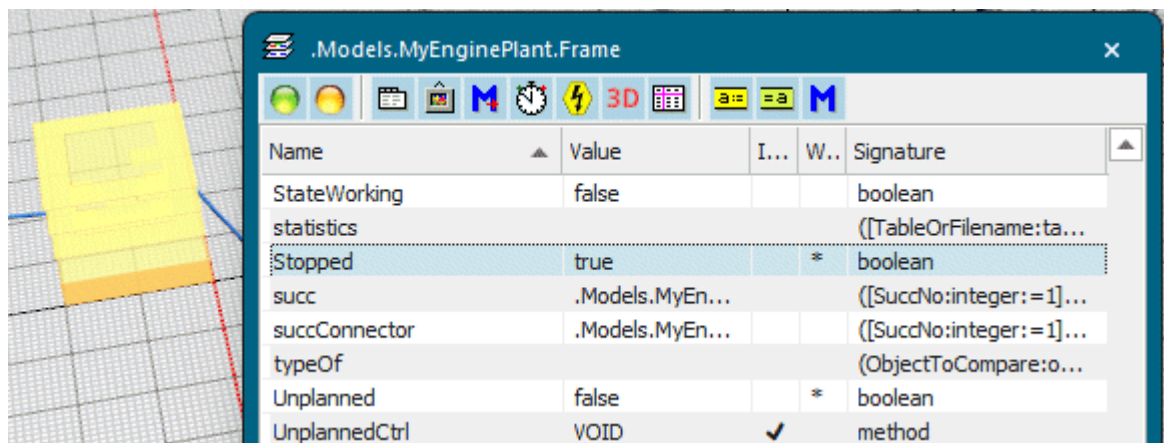
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Stopped Material Flow Objects and Frames

The *LockoutZone* stops the assigned material flow objects by setting their attribute *Stopped* to *true*.



The *LockoutZone* stops an assigned *Frame* by setting its attribute **Stopped** to true.



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